

Foundation (Jalapeno)

Q1. What is the vertex of the parabola $y = x^2 + 4x + 4$?

- (A) $(-2, 0)$
- (B) $(-2, 4)$
- (C) $(0, 4)$
- (D) $(-4, 0)$
- (E) $(-4, 4)$

Q2. Which equation represents a parabola with axis of symmetry $x = -3$?

- (A) $y = x^2 + 6x + 9$
- (B) $y = x^2 - 6x + 9$
- (C) $y = x^2 + 3x - 2$
- (D) $y = x^2 - 3x - 2$
- (E) $y = x^2 + 2x + 1$

Q3. What is the y -intercept of the parabola $y = x^2 - 4x - 3$?

- (A) -3
- (B) 3
- (C) 0
- (D) 4
- (E) -1

Q4. Which parabola opens downward?

- (A) $y = x^2 + 2x + 1$
- (B) $y = -x^2 + 2x - 1$
- (C) $y = x^2 - 2x - 1$
- (D) $y = -x^2 - 2x + 1$
- (E) $y = x^2 + 1$

Q5. What is the minimum value of $y = x^2 + 2x + 1$?

- (A) 0
- (B) 1
- (C) 2
- (D) -1
- (E) 3

Q6. What is the equation of the axis of symmetry for $y = x^2 + 4x + 4$?

- (A) $x = -2$
- (B) $x = 2$
- (C) $x = -1$
- (D) $y = -2$
- (E) $y = 2$

Q7. What are the x -intercepts of $y = x^2 - 4$?

- (A) $(-2, 0)$ and $(2, 0)$
- (B) $(-1, 0)$ and $(1, 0)$
- (C) $(-4, 0)$ and $(4, 0)$
- (D) $(-3, 0)$ and $(3, 0)$
- (E) $(-5, 0)$ and $(5, 0)$

Q8. Which equation represents a parabola with a maximum value?

- (A) $y = x^2 + 2x + 1$
- (B) $y = -x^2 + 2x - 1$
- (C) $y = x^2 - 2x - 1$
- (D) $y = -x^2 - 2x + 1$
- (E) $y = x^2 + 1$

Open Ended Questions (FRQ)**Q9.** Graph the parabola $y = x^2 - 2x - 3$ and identify its vertex, axis of symmetry, and x - and y -intercepts.**Q10.** Find the maximum or minimum value of $y = -x^2 + 4x - 3$ and explain your reasoning.

Chef's Tips

- Tip 1: Ensure students understand the properties of parabolas.
- Tip 2: Provide examples of different types of parabolas.
- Tip 3: Encourage students to use graphing tools to visualize parabolas.